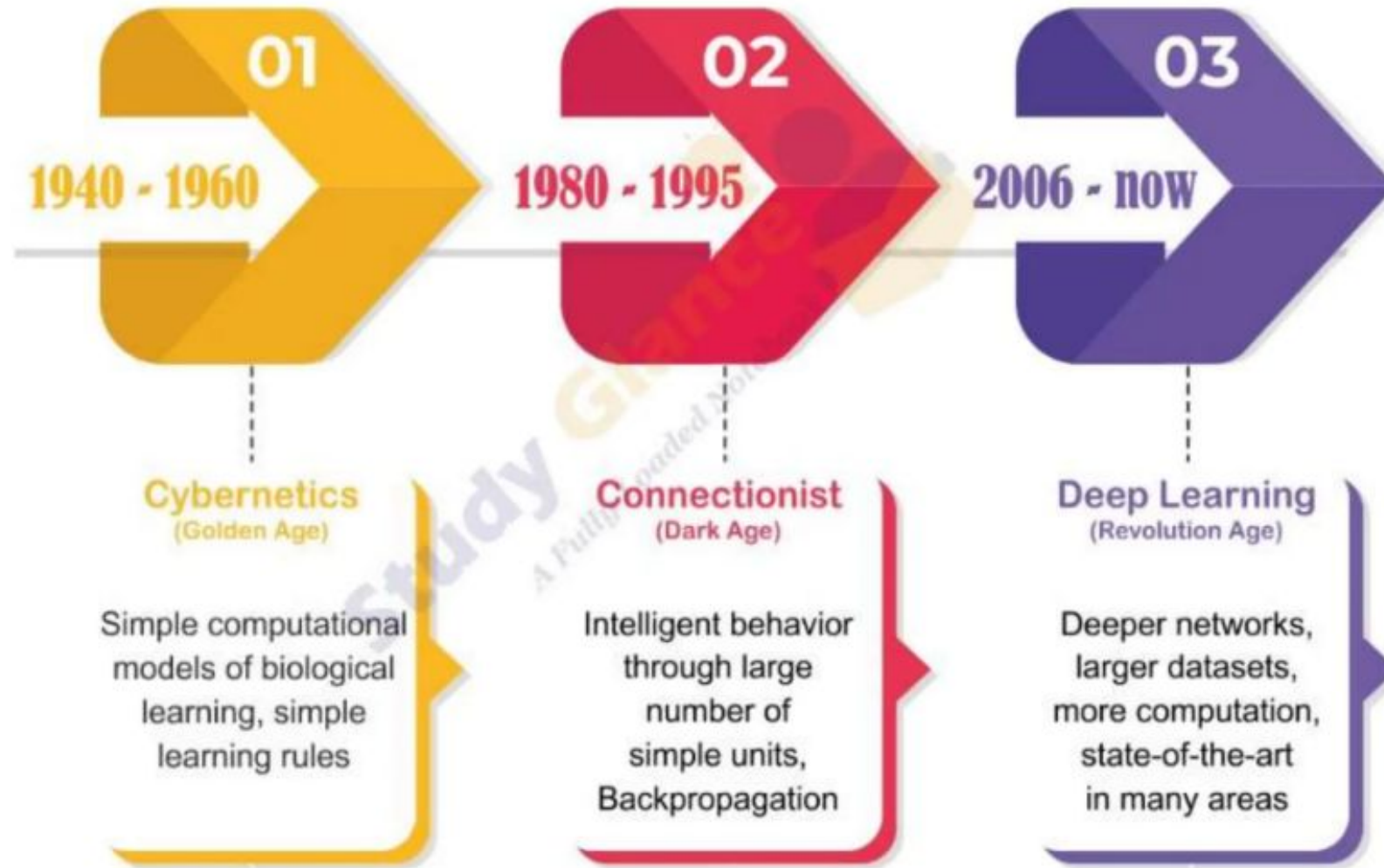


History trends of Deep Learning

Historical Trends in Deep Learning

- Deep Learning have been **three** waves of development:
- The first wave started with cybernetics in the **1940s-1960s**, with the development of theories of biological learning and implementations of the first models such as the perceptron allowing the training of a single neuron.
- The second wave started with the connectionist approach of the 1980-1995 period, with back-propagation to train a neural network with one or two hidden layers.
- The current and third wave, deep learning, started around 2006.

Three waves of development



- **Early Beginnings (1940s-1960s)**
- **1943: McCulloch-Pitts Neuron:** Conceptual foundation of artificial neurons.
- **1957: Perceptron:** Frank Rosenblatt introduces the first neural network model for binary classification.
- **1960: First Backpropagation Model:** Henry J. Kelley proposes early backpropagation concepts.

- **Rise of Neural Networks (1960s-1980s)**
- **1965: Multilayer Networks:** Ivakhnenko and Lapa develop early deep networks.
- **1980: Neocognitron:** Fukushima creates the first convolutional neural network (CNN).
- **1982: Hopfield Network:** John Hopfield introduces a recurrent neural network.
- **1985: Boltzmann Machine:** Hinton and Sejnowski's stochastic neural network.

- **Backpropagation and Revival (1980s-1990s)**
- **1986: Backpropagation Algorithm:** Popularized by Rumelhart, Hinton, and Williams, enabling deep neural network training.
- **1989: LeNet:** Yann LeCun's CNN for handwritten digit recognition.
- **Advances in Deep Learning (1990s-2000s)**
- **1997: LSTM Networks:** Hochreiter and Schmidhuber solve long-term dependency issues in RNNs.
- **2006: Deep Belief Networks:** Hinton et al. introduce efficient unsupervised pre-training for deep networks.

- **Modern Deep Learning Era (2010s-Present)**
- **2012: AlexNet:** Krizhevsky et al. win ImageNet competition, sparking deep learning boom.
- **2016: AlphaGo:** DeepMind's reinforcement learning model beats a world champion Go player.
- **2017: Transformer Models:** Vaswani et al. introduce the Transformer, revolutionizing NLP.
- **2020: GPT-3:** OpenAI's large-scale language model shows remarkable few-shot learning.
- **2020-Present: Vision Transformers (ViTs):** Transformers applied to vision tasks achieve state-of-the-art results.